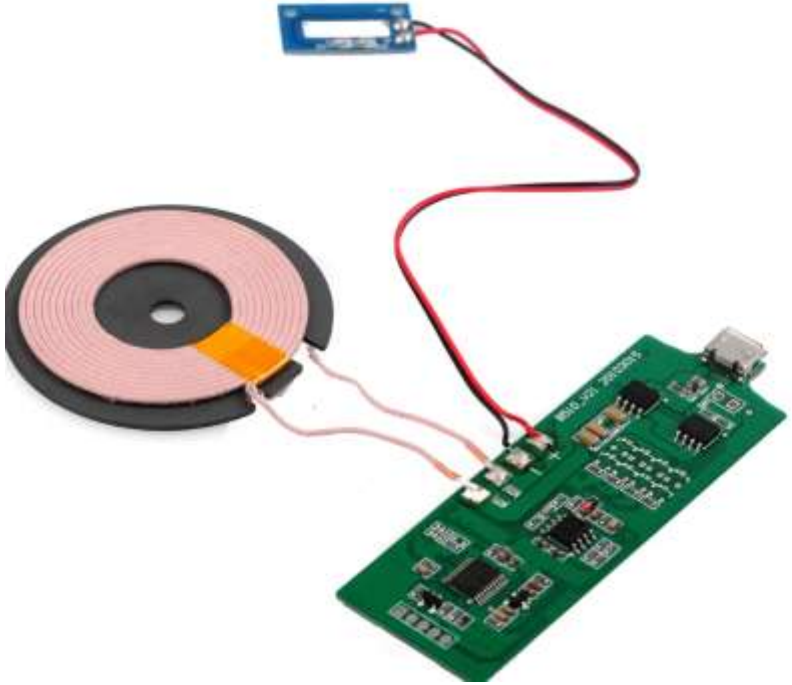


Preseason Intro

Groups

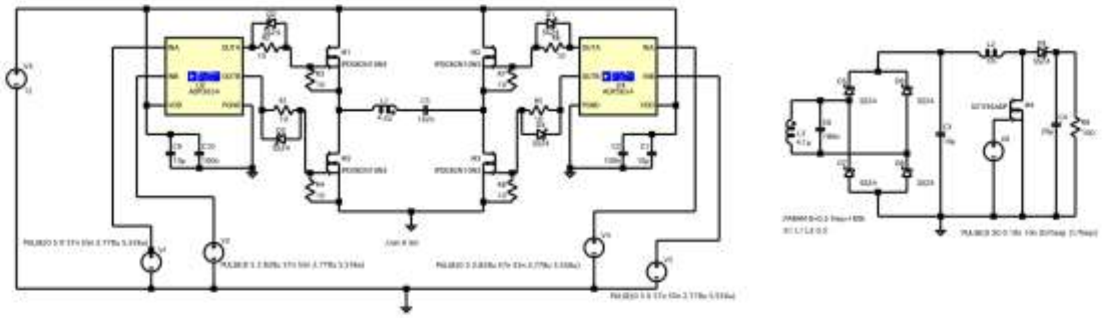
- Testing
- TET Circuit
- Motor
- Control
- Batteries

Testing

- ✓ Constructed transmitter and receiver coils and arranged them in planar shapes using a 3D-printed mold
- ✓ Created 10mm artificial human skin using silicone and a mold
- ✓ Done testing on the coils using oscillator and LCR meter
- ✓ Extracted resistance, coupling coefficient, and inductance data of the coil

- Develop a more detailed and thoughtful plan for coil testing, and repeat the it three times for validation
- Measure the temperature of the coils while inducting using a thermometer
- Analyse and develop solutions if temperature surpass safety limit

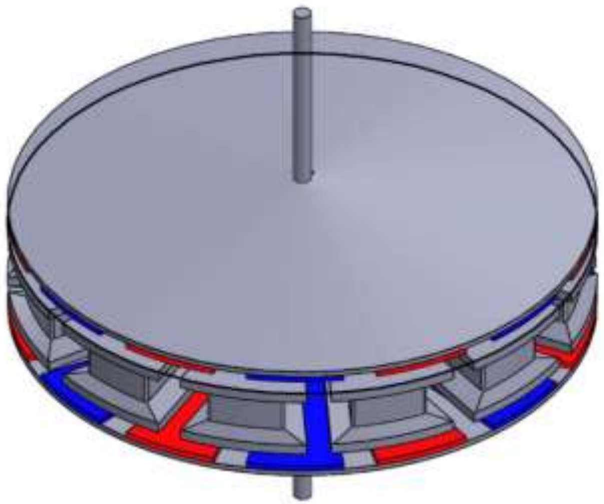
TET Circuit



- ✓ Developed inverter, rectifier, and boost converter circuit on LTSpice that meets the voltage and current requirement
- ✓ Converted the circuit schematics to PCB on KiCAD

- ○ Calculate the total circuit resistance and corresponding power loss; re-evaluate component choices and, where possible, substitute parts with lower series resistance while maintaining circuit performance.
- ○ Refine PCB design with a focus of footprint and power loss minimization

Motor



- ✓ Built basic CAD for the YASA motor and initiated magnetic ANSYS analysis on it

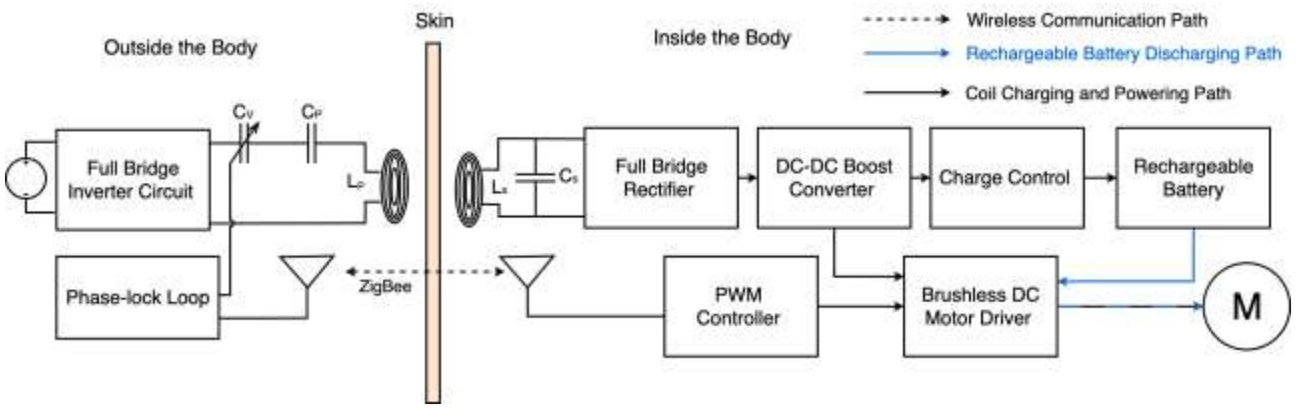
- ○ Discuss the purpose of the ANSYS analysis, as well as the motor's power requirement
- ○ Select motor material and refine CAD design with a focus on size and weight minimisation

Batteries

- ✓ Found lithium batteries that meet the required capacity requirement for internal battery

- ○ Continue search into small-sized rechargeable batteries that meets the capacity requirement (15Wh)
- ○ Determine types of external power source supply and the input requirement for the external circuit

Control



- ✓ Created block diagram showing the position of the frequency, battery, motor, and communication control

- Research into the construction of PLL automatic frequency tuning control or better alternative
- After finalising the battery choice, find the corresponding battery controller
- Select suitable motor drive control for the customized motor
- Research into the construction of the internal and external communication (Zigbee or better alternative)

Thank you for your effort!